



Hybrid CNN-BiLSTM model with HHO feature selection for enhanced fake news detection

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Abstract

Fake news, deliberately incorrect or misleading information presented as news, spreads through diverse platforms like online forums and traditional media. Its potential to harm lives, stir panic, and disrupt democratic systems makes it a pressing issue. Detecting fake news automatically is crucial to mitigate its impact. However, previous research often struggles with feature redundancy, suboptimal feature selection, and inefficient classification models, leading to lower accuracy and poor generalizability across datasets. To overcome these limitations, the present work introduces a methodology consisting of four distinct phases to detect fake news involving data preparation, feature extraction, feature selection, and classification. Data preprocessing eliminates ambiguities from the considered datasets and Linguistic Features (LFs) are extracted during the feature extraction phase to further perform the feature selection using Harris Hawks Optimization (HHO) algorithm. Furthermore, the CNN-BiLSTM hybrid model is then employed for classification, leveraging CNN's spatial feature extraction and BiLSTM's sequential learning capabilities for improved accuracy. In this work, we have performed the experiments on four publicly available datasets named ISOT, Kaggle, ConFake, and McIntire, where we achieved an accuracy of 98.89%, 98.25%, 98.56%, and 90.26%, respectively. Remarkably, this approach surpasses state-of-the-art methods by achieving an improved accuracy utilizing the HHO feature selection and CNN-BiLSTM hybrid model for fake news detection.

Keywords Fake news · Machine learning · HHO · Deep learning

1 Introduction

The widespread availability of digital media has significantly transformed contemporary society by facilitating communication, enhancing efficiency, and broadening access to knowledge. However, it has also amplified the spread of fake news, with severe consequences for many sectors of society. Notable examples include the Pizzagate incident (Hauck 2017) and mob lynchings in India (Mishra 2019).

Additionally, the dissemination of false information has led to harmful outcomes such as alterations in health behaviour intentions (Greene and Murphy 2021), increased vaccine hesitancy (Islam et al. 2021), and substantial economic losses (Brown 2019). Fake news encompasses various forms of misinformation that can undermine the transmission of accurate knowledge. Misleading headlines use sensational or deceptive titles that differ from the actual content. Fabricated stories are entirely false narratives created to mislead readers or promote specific agendas. Manipulated content involves altering photographs, videos, or quotations to misrepresent events or individuals. Conspiracy theories (Zhou et al. 2020) propagate erroneous and often convoluted ideas about hidden agendas or intentions. Satire or parody, although intended to be humorous, can sometimes be misinterpreted as genuine news due to a lack of context. Clickbait relies on sensationalized content to attract attention, often leading to unreliable information. Propaganda spreads biased or misleading information to achieve specific political, social, or ideological goals. False statistics or facts involve the use of inaccurate or manipulated data to support

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a particular narrative. Impersonation, also known as spoofing, involves creating fake accounts or websites that mimic authentic sources. Lastly, out-of-context information arises when data or statements are presented without proper context, leading to incorrect interpretations. Understanding and recognizing these various forms of fake news is essential for developing critical thinking skills and effectively evaluating information sources.

To combat the effects of fake news, several straightforward strategies can be adopted. First, ensure that the news comes from reputable and reliable sources. Second, verify the information using credible fact-checking websites or trustworthy references. Third, cultivate critical thinking skills by recognizing bias and sensationalism in news and considering multiple perspectives. To gain a balanced understanding, diversify your sources by reading news from various outlets. Fifth, exercise caution when consuming news on social media platforms, where false information can spread rapidly. Before sharing any news, confirm its accuracy to prevent the unintentional dissemination of fake news. Sixth, report fake news to the relevant platform or website to help curb its spread. Finally, support quality journalism by contributing to credible news organizations that uphold rigorous journalistic standards. By following these steps, individuals can navigate the information landscape more effectively and mitigate the impact of fake news. Various Machine Learning (ML) (Gravanis et al. 2019; Verma et al. 2021; Jain et al. 2023, 2020; Maan et al. 2022; Jain et al. 2022) and Deep Learning (DL) (Choudhary and Arora 2021) techniques have been extensively studied for detecting fake news. In the field of DL, models such as Convolutional Neural Network (CNN) (Verma et al. 2021), Long Short-Term Memory (LSTM) networks (Choudhary and Arora 2021; Dixit et al. 2022), and more complex hybrid architectures (Choudhary et al. 2021; Varshini et al. 2023; Nithya and Sahayadhas 2023; Okunoye and Ibor 2022) have been developed. These models demonstrate strong capabilities in analyzing news content and effectively distinguishing between real and fake news.

Detecting and combating fake news using the most effective DL models is essential for preserving trustworthy information and shielding individuals from fake news. However, existing approaches face several limitations. These models often struggle to address evolving techniques of fake news and lack adaptability to diverse domains such as healthcare, entertainment, or education. Additionally, challenges like dataset biases and privacy concerns affect their fairness and transparency.

Many previous studies (Kaliyar et al. 2020; Choudhary and Arora 2021; Pérez-Rosas et al. 2017) have relied on single datasets, limiting their generalizability across different domains. Others (Alghamdi et al. 2023; Zivkovic et al. 2021; Jin et al. 2016; Shah and Kobti 2020) have used short-text

datasets, which fail to capture complex linguistic structures inherent in fake news articles. Moreover, ML-based models (Pérez-Rosas et al. 2017; Zhou et al. 2004; Ahmed et al. 2018) incorporating linguistic features (LFs) often fail to achieve optimal classification accuracy, while DL-based models (Zhou et al. 2020; Varshini et al. 2023), despite automating feature extraction, struggle with performance enhancement due to the absence of robust feature selection mechanisms. Some studies (Yang et al. 2018; Reddy et al. 2020) have attempted to combine linguistic and latent features but lacked effective feature selection techniques, leading to redundant or irrelevant features being included in the classification process. Furthermore, optimization techniques have primarily been applied to hyperparameter tuning (Dixit et al. 2022; Zivkovic et al. 2021) rather than feature selection, overlooking the potential impact of selecting the most relevant LFs.

The proposed study introduces a novel framework combining the HHO algorithm with a hybrid CNN-BiLSTM (Bidirectional Long Short-Term Memory) architecture for fake news detection. The HHO algorithm, inspired by the cooperative and predatory behavior of Harris hawks, optimizes feature selection by dynamically identifying the most relevant patterns in complex datasets, thereby enhancing model performance (Heidari et al. 2019; Alabool et al. 2021). Complementing this, the hybrid CNN-BiLSTM model leverages CNN's ability to extract local features and BiLSTM's capacity to capture contextual and sequential dependencies, creating a comprehensive approach to text analysis. Unlike existing DL models, which often rely on static feature sets or singular architectures, this framework dynamically adapts to data characteristics through HHO-driven feature selection while combining CNN and BiLSTM for enhanced interpretability and accuracy. Additionally, unlike traditional optimization techniques that may converge prematurely or struggle with high-dimensional data, HHO offers a more flexible and effective solution by mimicking the adaptive hunting strategies of hawks (Alabool et al. 2021; Heidari et al. 2019).

To address existing limitations, our approach integrates an optimization technique for selecting optimal features from a comprehensive linguistic feature set. This set includes features commonly used in fake news detection, hate speech detection, rumour detection, fake review detection, and satire detection. By leveraging HHO for feature selection and employing a CNN-BiLSTM model for classification, our approach enhances efficiency and accuracy in fake news detection. This raises important Research Questions (RQ):

RQ 1: How does the feature selection process using HHO enhance the performance of fake news detection?

RQ 2: Will integrating the HHO method with CNN-BiLSTM yield superior accuracy results in fake news detection?

RQ 3: Does the HHO-CNN-BiLSTM approach surpass existing fake news detection systems in terms of performance?

Additionally, the study includes a comparison of ML models to identify the most accurate model for text processing while minimizing the number of parameters to enhance execution performance. The proposed approach is evaluated on multiple datasets, including ISOT, Kaggle, McIntire, and ConFake, to assess its efficacy and robustness. The primary contribution of this study can be succinctly summarized as follows:

1. We propose a novel fake news detection model, HHO-CNN-BiLSTM, which integrates the HHO algorithm for feature selection with a CNN-BiLSTM architecture for classification. This combination improves both feature selection efficiency and classification accuracy.
2. We extract eighty LFs from the text and use the HHO algorithm to select the most relevant features. This reduces redundancy and enhances model performance.
3. The selected features are combined with word embedding and then passed to the CNN-BiLSTM model, where CNN captures spatial patterns, and BiLSTM learns sequential dependencies. This integration helps accurately detect fake news while adapting to evolving fake news patterns.

This article follows a well-structured approach. Section 1 provides context and background, while Sect. 2 reviews relevant literature. Section 3 presents the proposed methodology, followed by Sect. 4, which details the evaluation criteria and comparative analysis with existing models. Additionally, the study assesses the efficiency of the proposed approach against recent methods. Finally, Sect. 5 concludes with insights and future research directions, contributing to advancements in fake news detection.

2 Literature survey

The effects of fake news on people and communities have made its detection and prevention a critical area of research. Various studies have proposed diverse approaches to tackle this challenge, focusing on LFs and ML-based methodologies. Reddy et al. (2020) employed stylometric and word vector features, such as Term Frequency (TF), Term Frequency and Inverse Document Frequency (TF-IDF), Continuous Bag of Words (CBOW), and Skip-gram, along

with 50 LFs, to detect fake news. Combining two datasets, McIntire and FakeNewsNet, and using classifiers such as Logistic Regression (LR), Support Vector Machine (SVM), Naïve Bayes (NB), Random Forest (RF), K-Nearest Neighbors (KNN), Gradient Boosting (GB), Adaboost, and Bagging, their method achieved a 95.49% accuracy. However, the study's limitations included a small dataset size and the overfitting of LFs with SVM. X. Zhou and colleagues Zhou et al. (2020) proposed a methodology for fake news detection using PolitiFact and BuzzFeed datasets. They extracted lexical, syntactic, semantic, and discourse-level features from news article text and fed them into ML models, either individually or in combination, including LR, NB, SVM, RF, and XGBoost. Their approach achieved 89.2% accuracy on the PolitiFact dataset and 87.9% on BuzzFeed. This study incorporated clickbait and disinformation-related features but was limited by a smaller corpus size and the presence of unused photos in the datasets. Pérez-Rosas et al. (2017) focused on main LFs such as n-grams, punctuation, psycholinguistics, readability, and syntax, achieving 76% accuracy on two datasets: FakeNewsAMT and Celebrity. These datasets cover seven domains, but the study faced challenges with low model accuracy and a limited number of features used for fake news detection. Gravanis et al. (2019) introduced a new unbiased dataset by combining articles from four sources: Kaggle-EXT, McIntire, BuzzFeed, and PolitiFact, resulting in 3,004 articles. After extracting 57 LFs and using Word Embedding (WoE), they applied ML classifiers such as SVM, NB, Decision Tree (DT), KNN, Adaboost, and Bagging. SVM yielded the best accuracy of 94.9%. Their methodology was further validated across individual datasets, achieving accuracies of 99.0% (Kaggle-EXT), 72.7% (BuzzFeed), 84.7% (PolitiFact), and 81% (McIntire). However, their study did not include user-based features. Khan et al. (2021) conducted experiments using ML and DL models on three distinct datasets. Similarly, Shu et al. (2019) used ML classifiers with features derived from article text, utilizing the PolitiFact and BuzzFeed datasets. Reis et al. (2019) experimented with the BuzzFeed dataset using t-SNE to detect fake news with linguistic, semantic, and credibility features. Finally, Verma et al. (2021) proposed the WELFake model, integrating LFs with WoE vector attributes. While their phased approach offered some insights, it did not yield significant improvements in results.

After applying ML and DL-based approaches, researchers have increasingly turned to metaheuristic algorithms for selecting optimal features or performing hyperparameter tuning to enhance model performance for classification tasks. Ozbay and Alatas (2019) innovatively employed Grey Wolf Optimization (GWO) and meta-heuristic algorithms, resulting in enhanced efficiency for social media data analysis. Kumar et al. (2019) leveraged Particle Swarm Optimization (PSO) to optimize feature sets extracted from

tweets, boosting classifier performance for rumor classification. Sahoo and Gupta (2021) introduced a method using Chrome's built-in features to detect false information on Facebook automatically. Jain et al. (2023) proposed "Con-Fake," an algorithm combining content-based and word vector features for fake news detection. Their model achieved an impressive accuracy of 97.31% on diverse datasets. Yu et al. (2017) criticized Recurrent Neural Network (RNN)-based algorithms and introduced CNN models to identify propaganda, while Shu et al. (2019) pragmatically combined co-attention techniques with DL for fake news detection. Li et al. (2021) developed unsupervised fake news detection by integrating news context and content data, enhancing fake news identification. Chen et al. (2018) employed an attention-based RNN for early rumour detection. Wang (2017) proposed a methodology using CNN and bidirectional recurrent transitional memories for DL. Yin et al. (2019) used SVM classifiers for pattern analysis, while Toumi and Bouramoul (2022) introduced an ensemble model combining CNN, LSTM, and Convolutional LSTM. Nasir et al. (2021) proposed a CNN/RNN hybrid approach, addressing the challenges of RNN optimization. Nithya and Sahayadhas (2023) presented Meta-heuristic Searched-Ensemble Learning (MS-EL) for fake news detection, achieving superior accuracy through advanced feature selection. Jing et al. (2023) proposed a progressive fusion network for multimodal disinformation detection, effectively fusing textual and visual information. Zhang et al. (2023) used event and topic extraction algorithms with a two-stage mechanism to optimize news data handling. Singh et al. (2021) combined textual and visual information using various ML techniques, achieving a predictive accuracy of 95.18%. These diverse strategies collectively contribute to the ongoing advancement of fake news detection methodologies. Giachanou et al. (2020) proposed a multimodal model using Bidirectional Encoder Representations from Transformers (BERT) for linguistic representations and a 16-layer VGG network for visual features, achieving 79.55% F1 score on FakeNewsNet data. Kirchknopf et al. (2021) classified fake news in the Fakeddit dataset using text, comments, photos, and more, with results accurate within 5%. Li et al. (2021) introduced the entity-oriented multimodal alignment and fusion network with accuracy levels of 92.3%, 97.4%, and 80.5% on Fakeddit, Weibo, and Twitter datasets. Khattar et al. (2019) utilized multimodal variational encoders for text and picture features, suggesting potential for complexity enhancement. Wang et al. (2018) proposed an Event Adversarial Neural Network (EANN) to extract features from text and images. Jin et al. (2017) used RNN and local attention mechanism, while Song et al. introduced the CARMN framework with CNN. Wu et al. (2021) introduced MCAN for disingenuous media detection. Xie et al. (2024) proposed Multiknowledge and Large Language Model (LLM)-inspired Heterogeneous

Graph Neural Network (GNN), combining LLMs, Knowledge Graphs (KGs), and heterogeneous GNNs for fake news detection using a graph transformer and metapath-guided aggregation. Tested on four datasets, it outperformed ten baselines in multiple metrics. However, the reliance on extensive KGs and LLMs may limit scalability to resource-constrained settings. Peng et al. (2024) developed a model, integrating local and global semantic features for multimodal fake news detection with robust contextual representations. It achieved a 2.5% accuracy improvement over ten baselines on two datasets. The approach, however, may struggle with datasets lacking diverse semantic contexts. Luvembe et al. (2024) introduced CAF-ODNN, addressing uncorrelated multimodal features using attention fusion and optimized Deep NNs, achieving superior results on four datasets. Its reliance on precise alignment and normalization could limit effectiveness on noisy or incomplete multimodal data.

After studying various approaches, we identified several limitations in previous works. Many researchers relied on a single dataset for their experiments, which is insufficient for robust analysis. Additionally, some utilized only short-text datasets, limiting the scope of their findings. Numerous studies employed ML models based on LFs for fake news detection but failed to achieve optimal results. While others adopted DL models to automate feature extraction, they often struggled to improve overall performance. A few approaches combined linguistic and latent features but failed to select the most effective features for the task. Optimization techniques for hyperparameter tuning were also applied, but primarily on latent features, overlooking the potential of LFs.

To address these issues, we adopted an optimization technique to select optimal features from a comprehensive linguistic feature set, which includes features commonly used for fake news detection, hate speech detection, rumour detection, fake review detection, and satire detection. After selecting the optimal features, we used a CNN-BiLSTM model for classification. The effectiveness of this approach is thoroughly analyzed in later sections, where its performance is compared against existing methods. Table 1 provides insights into diverse approaches across domains for fake news identification, highlighting key research gaps and potential improvements in detection strategies.

3 Proposed methodology

Fake news detection is the process of assessing the authenticity of news articles by determining whether the information they contain is true or false. In this study, fake news detection is framed as a binary classification problem, where the objective is to classify a given news article N into one of two classes: C_1 (real news) or C_2 (fake news). Formally,

Table 1 Summary of fake news detection methods

Reference(s)	Dataset(s)	Dataset size	Input	Technique	Feature extractor	Accuracy	Optimization used	Limitations
Mallik and Kumar (2024)	TI-CNN dataset, Kaggle, SMS Spam, and ISOT	20,015, 20,800, 5574, 44,898	Text	Hybrid Word2vec and LSTM model	Word2vec	92%, 97.5%, 98%, 97%	No	Not utilized the LFs
Zhou and Cai (2024)	NewsHeadline, COVID, and PolitiFact	21,367, 6420, 17,054	Text	Multi-scale recurrent CNN	Latent Features Via Electra	91.63%, 95.96%, 79.63%	No	Lack of an intelligent focal loss method, as determining its optimal hyperparameters is time-intensive
Varshini et al. (2023)	Constraint AAI-Shared Task, COVID Infodemic, COVID Fake News, and Multilingual COVID	10,700, 260, 4000, 2670	Text	RDGT-GAN	GAN	83.65%, 76.92%, 82%, 84%	Yes	The small size of all datasets leads to limited performance
Alghamdi et al. (2023)	COVID-19 fake news	6420	Text	CT-BERT + BiGRU	Latent features using CNN and BiGRU	98.46%	No	Not include the LFs and depend on single dataset
Dixit et al. (2022)	BuzzFeed, GossipCop, ISOT, PolitiFact	1700, 17,520, 44,898, 700	Text	PPCA, Levy Flight based LSTM	Latent Features using LSTM	91%, 94%, 89%, 90%	Yes	LFs were not utilized, resulting in suboptimal model accuracy
Garg and Sharma (2022)	BuzzFeed, McIntire, Reuter, and PolitiFact	182, 6336, 44,898, 240	Text	LR	LIWC, TF, and TF-IDF	90%, 86.9%, 90.8%, and 93.8%	No	Experiment performed on a small size dataset and limited features incorporated
Verma et al. (2021)	WELFake	72,134	Text	SVM, KNN, NB, DT, Bagging, and Adaboost	LIWC, Textstat	96.73%	No	A random feature selection approach was used
Zivkovic et al. (2021)	COVID-19	10,700	Text	KNN with AOA	BOW, TF, TF-IDF	75%	Yes	A small dataset was used, resulting in poor performance
Kaliyar et al. (2021)	Fake news dataset	20,800	Text	FakeBERT, CNN, LSTM	BERT	98.90%	No	Not include the LFs and depend on single dataset
Ghanem et al. (2020)	News articles and Twitter dataset for different categories (Propaganda, hoax, clickbait, and satire)	8529	Text	Emotion Infused Neural Network	LIWC and Neural Network	Different for each dataset, max: 96% on Clickbait	No	Different length of articles used

Table 1 (continued)

Reference(s)	Dataset(s)	Dataset size	Input	Technique	Feature extractor	Accuracy	Optimization used	Limitations
Zhou et al. (2020)	PolitiFact, Gossip-Cop	1056, 22,140	Text and Images	CNN, VGG-19, image2sentence model, cross-modal relationship, SAFE	Latent Features using CNN	87.4%, 83.8%	Yes	Only latent features were used, leading to suboptimal performance
Shah and Kobti (2020)	Twitter, Weibo	13,924, 9528	Text and Images	Sentiment analysis, Image segmentation, Cultural algorithm optimization	Visual and Sentiment features (Self Implemented)	79.8%, 89.1%	Yes	Short text utilized, resulting in poor accuracy
Choudhary and Arora (2021)	BuzzFeed Political news, and Random Political news	101, 150	Text	LFs based sequential LSTM model	LIWC, Textstat, and Text polarity	84.52%, 80.5%	No	Small datasets used, leading to poor model performance

given a dataset D consisting of news articles N_i with corresponding labels $y_i \in \{C_1, C_2\}$, the goal is to learn a mapping function $f : N \rightarrow y$ that minimizes the classification error while ensuring robustness and scalability (Yang et al. 2018).

The problem addressed in this study focuses on designing an effective computational model to detect and classify fake news articles, thereby mitigating the impact of fake news on society. The present work introduces a methodology comprising four distinct phases for detecting fake news including data preparation, feature extraction, feature selection, and classification. Data preprocessing eliminates ambiguities from textual content and LFs are extracted during the feature extraction phase. Feature selection uses the HHO algorithm to retain the most relevant features. The optimal features are then combined with word embedding to fed into the CNN-BiLSTM hybrid model for classification considering four datasets including ISOT, Kaggle, ConFake, and McIntire, demonstrates the robustness of the proposed methodology.

3.1 Dataset description

Within this investigation, we harnessed four distinct datasets of varying sizes to validate the efficacy of the proposed model. Comprehensive details about the employed datasets are encapsulated within Table 2. This tabulated information offers insights into the foundational components underpinning our study's evaluation.

3.1.1 ConFake dataset

The ConFake dataset, introduced by Jain et al. (2023), is designed to mitigate overfitting in ML models and enhance training effectiveness. It consolidates data from five diverse sources: Kaggle, Reuter's, McIntire's, BuzzFeed's, and PolitiFact's. Comprising a total of 72,455 news articles, the dataset encompasses 35,185 verified true news articles and 37,270 fake news articles.

3.1.2 ISOT dataset

The University of Victoria's ISOT lab has compiled the ISOT (Ahmed et al. 2018) as a dataset for detecting fake news. It has 45,000 different news articles. Title, date, subject, and article body are all included for each piece of news covering politics and the globe from 2016 and 2017. Truthful reports come from Reuters, while those with dubious veracity come from sites like PolitiFact and even Wikipedia. The proportion of fake to true stories in this dataset is about equal; hence, it is likewise regarded as well-balanced.

Table 2 Dataset description

Dataset	#Articles	#True articles	#Fake articles
ConFake (Jain et al. 2023)	72,455	35,185	37,270
Kaggle (Fake News Kaggle Dataset 2020)	20,800	10,387	10,413
ISOT (Ahmed et al. 2018)	44,898	21,416	23,482
McIntire (McIntire Dataset 2020)	6335	3171	3164

3.1.3 McIntire

The McIntire Dataset (McIntire Dataset 2020), curated by George McIntire, offers a substantial collection of genuine and fabricated news items. Primarily centred around the political landscape of the 2016 US elections, the dataset consists of a total of 6335 news stories. Among these, there are 3171 verified real news stories and 3164 fake news stories. The dataset predominantly centres on US political affairs.

3.1.4 Kaggle

The dataset (Fake News Kaggle Dataset 2020) comprises a collection of 20,800 articles, encompassing both authentic and fabricated news, sourced from a variety of real-world outlets. Among its contents, there are 10,387 articles classified as true news and 10,413 articles classified as fake news. Each of these datasets serves as a valuable resource for training and evaluating ML models aimed at detecting and classifying fake news in different contexts.

3.2 Data preprocessing

Data is a critical component in any ML model, and effective data preprocessing is essential for optimizing model performance. In this study, we employ a two-phase data preprocessing approach tailored to extract LFs and word embeddings.

In the first phase, preprocessing is conducted to prepare the data for linguistic feature extraction through the following steps:

- i. Removing duplicate articles to eliminate redundancy.
- ii. Handling missing values to ensure data completeness.
- iii. Manually filtering out irrelevant articles (e.g., "Error 404: Page Not Found") that do not contribute to fake news detection.
- iv. Applying stemming and lemmatization to normalize text while preserving meaning.

At the end of this phase, the cleaned textual data is ready for linguistic feature extraction. In the second phase,

preprocessing is performed for word embedding extraction with additional refinements:

- i. Steps (i)–(iii) are repeated as in the first phase.
- ii. Convert the text into lowercase.
- iii. Removing punctuation marks, special characters, and numbers, including question marks, exclamation marks, ellipsis, hashtags, and "@" symbols, as they hold semantic significance in detecting fake news.
- iv. Applying stemming and lemmatization to standardize the text format.

Once preprocessing is complete, the refined textual content is ready for feature extraction, where LFs and word vector representations are generated for model training. This structured approach ensures that the dataset is clean, consistent, and optimized for effective fake news detection.

3.3 Feature extraction

After preprocessing the textual data, we extracted 80 LFs to capture essential textual characteristics for fake news detection. Table 3 shows the LFs and their corresponding extraction tools. These features were selected based on their effectiveness in prior studies (Gravanis et al. 2019; Hamed et al. 2023; Verma et al. 2021; Choudhary and Arora 2021) and their relevance to linguistic, syntactic, and semantic patterns. The feature extraction process was performed in multiple steps using a combination of self-programmed Python scripts, the LIWC (Linguistic Inquiry and Word Count) dictionary (Pennebaker et al. 2001), NLTK, TextStat, and TextBlob libraries.

First, quantity-based features such as character count, word count, sentence count, and paragraph count were computed to analyze the structural complexity of news articles. Complexity-based features, including average word length, average sentence length, and average syllables per word, were then extracted to measure linguistic richness. Next, uncertainty features were identified by detecting modal verbs, tentative words, question marks, numbers, and ellipsis, which often indicate fake news. To assess subjectivity, we extracted features related to factual and subjective verbs, as well as motion verbs, greeting words, and reporting verbs. Additionally, sentiment features such as positive words, negative words, anger-related

Table 3 Linguistic features and their corresponding extraction tools

Category	Features	Tool used
Quantity	#Characters, #Words, #Sentences, #Paragraphs	Self-Programmed in Python
	#Syllables, #Articles, #Verbs, #Adverbs, #Adjectives	LIWC Dictionary
	#Stopwords	NLTK Library
	#Uppercase Letters, #Self-referencing, #Group-referencing	Self-Programmed in Python
Complexity	Avg. Words per Sentence, Avg. Characters per Word, Avg. Syllables per Word, Avg. Sentences per Paragraph	Self-Programmed in Python
Uncertainty	Certainty, Generalizing Terms, Tentative Terms	LIWC Dictionary
	Numbers, Question Marks, Ellipsis (...)	Self-Programmed in Python
Subjectivity	Factual Verbs, Motion Verbs, Subjective Verbs	LIWC Dictionary
	Greeting Verbs, Reporting Verbs	Self-Programmed in Python
Non-immediacy	#First-person Pronouns, #Second-person Pronouns, #Third-person Pronouns	LIWC Dictionary
Sentiment	Positive Words, Negative Words, Exclamation Marks, Anger Words, Anxiety Words, Sadness Words	LIWC Dictionary
Diversity	Content Words, Function Words, Pronouns, Prepositions, Determiners, Conjunctions	LIWC Dictionary
	Lexical Diversity, #Unique Words (Types), #Tokens, #Polysyllables	Self-Programmed in Python
Informality	Swear Words, Netspeak Words, Assents, Non-fluencies, Fillers	LIWC Dictionary
Specificity	Temporal Words, Spatial Words, Causation Terms, Exclusive Terms, Cognitive Processes, Perceptual Processes	LIWC Dictionary
Readability	Flesch Reading Ease Index, Flesch-Kincaid Grade Level, Automated Readability Index, Coleman Liau Formula	TextStat Python Library
	SMOG Index, Gunning Fog Index, New Dale-Chall Formula, Linsear Write Formula	
Writing Pattern	#Question Marks (?), #Exclamation Marks (!), #Single Quotes (‘ ’), #Double Quotes (“ ”), #Hashtags (#), #At Symbols (@)	Self-Programmed in Python
	#Big Words (> 15 characters), #Short Words (< 4 characters)	Self-Programmed in Python
Psycho-linguistic	Text Polarity, Subjectivity	TextBlob Python Library

terms, and exclamation marks were measured using the LIWC dictionary. Diversity features like lexical richness, unique words, function words, and pronoun usage were computed to understand writing style variations.

Furthermore, we extracted informality features (swear words, netspeak, non-fluencies, fillers) and specificity features (causation terms, cognitive processes, perceptual words) to identify patterns unique to deceptive news articles. Readability metrics, including Flesch Reading Ease, Gunning Fog Index, and Automated Readability Index, were computed using the TextStat library to evaluate textual complexity. Lastly, writing pattern features, such as special character frequency (e.g., ?, !, #, @) and psycho-linguistic attributes (text polarity and subjectivity), were incorporated to enhance the model's ability to differentiate between real and fake news. By systematically extracting and structuring these LFs, we ensured a comprehensive feature set that captures various stylistic, structural, and contextual aspects of the text. These features were then fed into the HHO algorithm for feature selection, allowing us to retain the most relevant attributes while eliminating redundancy.

In Table 4, the ConFake dataset, which combines five datasets, was selected to demonstrate the significance of

Table 4 Analysis of key linguistic features in the ConFake dataset

Features	True news	Fake news
Average #capital letters	115.60	119.03
Average #quoted content	1.85	1.48
Average #URLs	0.0019	0.2046
Average #tokens	665.82	583.97
Average #types	294.79	270.73
Average size of words	3.94	3.98
Average type-token ratio	0.033	0.047
Average #sentences	25.16	20.27
Average size of sentence	28.96	32.05
Average #content words (noun, adverbs, adjective, etc.)	174.99	155.15
Average #first pronoun	3.72	2.58
Average #second pronoun	2.47	1.19
Average #third pronoun	15.71	17.53
Average #positive sentences	2.01	2.05
Average number of question marks (?)	0.52	1.37
Average number of exclamation marks (!)	0.15	0.73
Average readability index	37.39	36.05

linguistic and structural features in fake news detection. Since it includes data from multiple sources, it provides a broader perspective on how these features help distinguish between fake and true news. The analysis shows that fake news articles are generally shorter, with fewer words (583.97 vs. 665.82 in true news) and sentences (20.27 vs. 25.16). However, fake news sentences are longer on average (32.05 words per sentence compared to 28.96 in true news), making them more complex. Fake news also includes more URLs (0.2046 vs. 0.0019), question marks (1.37 vs. 0.52), and exclamation marks (0.73 vs. 0.15), which suggests a more dramatic or exaggerated writing style. Additionally, fake news uses fewer content words (nouns, adjectives, adverbs) and first- and second-person pronouns, but third-person pronouns appear more frequently. True news contains more quoted content (1.85 vs. 1.48), which may indicate a greater reliance on verifiable sources. The readability scores are similar, but fake news is slightly harder to read (36.05 vs. 37.39). This analysis highlights clear differences in writing style, sentence structure, and the way information is presented in fake and true news. These features play an important role in detecting fake news.

3.4 Harris Hawk optimization

Following data preprocessing, the retrieved characteristics undergo a selection process that is aided by the HHO. During this step, the HHO algorithm effectively discerns the most significant characteristics, which are subsequently directed into a CNN. The HHO, an innovative accomplishment by Heidari et al. (2019), represents a notable progression in the field of swarm intelligence optimization algorithms. This algorithm emulates the watchful hunting behavior of hawks, drawing inspiration from their methodical approach to locating prey. The method entails a cooperative endeavor among hawks, wherein they pool their perspectives to efficiently accomplish the task of hunting (Heidari et al. 2019).

The HHO method comprises two unique phases, namely diversification, and intensification, which correspond to certain stages of development. The initial stage, characterized by the processes of diversification and the subsequent shift towards intensification, bears a striking resemblance to the systematic strategy employed by predators while pursuing their prey. During the intensification stage, a range of methods are utilized, including soft besiege, hard besiege, soft besiege with progressive swift pounce, and hard besiege with progressive rapid pounce. The selection of options is influenced by the prevailing situations, so ensuring the ability to react to the circumstances at hand.

3.4.1 Diversification

To effectively identify and determine the position of prey, individuals engage in a persistent exploration of the barren surroundings. Harris hawks employ two separate hunting strategies while perched in various locales. The initial method, as depicted in Eq. 1 for values of d less than 0.5, involves the influence of other members within the family and the rabbit's location on the behavior of the perch. The second strategy, as illustrated in Equation 1 when the value of d is greater than or equal to 0.5, portrays Harris hawks positioned on various lofty trees.

$$\partial(t+1) = \begin{cases} \partial_{rand}(t) - p_1 \|\partial_{rand}(t) - 2p_2\partial(t)\| & \text{if } d \geq 0.5 \\ (\partial_{rabbit}(t) - \partial_m(t)) - p_3(LB + p_4(UB - LB)) & \text{if } d < 0.5 \end{cases} \quad (1)$$

$$\partial_m(t) = \frac{1}{N} \sum_{i=1}^N \partial_i(t) \quad (2)$$

Where $\partial(t+1)$ depicts the location of hawks in the following generation, ∂_{rabbit} represents the location of rabbit, $\partial(t)$ is the current generation vector, p_1, p_2, p_3, p_4 & d are different random numbers obtained from 0 to 1, p_5 is a dynamically chosen hawk, ∂_{rand} is a variable, $\partial_m(t)$ stands for the position of all the hawks as a whole on average.

3.4.2 Switching from diversification to intensification

The initial focus of this study involves the examination of diversity inside the search space, followed by the subsequent utilization of optimization strategies to exploit the surrounding region of the solution, commonly referred to as intensification. The HHO model incorporates the escape Energy (E) of prey as a means of simulating the behavior observed in rabbits. The diversification process is chosen when the cardinality of set E is greater than or equal to 1, whereas the intensification process is employed when the cardinality of set E is less than 1. Consequently, the transition from diversification to intensification is achieved by harnessing the vanishing energy of prey. The estimation of energy is stated in Eq. 3, as depicted below.

$$E = 3E_o \left(1 - \frac{t}{T}\right) \quad (3)$$

3.4.3 Intensification

The method consists of four strategies: soft besiege (a), hard besiege (b), soft besiege with progressive quick pounce (c), and hard besiege with progressive quick bounce (d). The subsequent section provides a more comprehensive explanation of each of these tactics.

(a) **Soft Besiege:** In a situation of mild siege, the intended targets possess the capability to flee, yet they find themselves incapable of doing so. When the value of p is more than or equal to 0.5, and the absolute value of E is greater than or equal to 0.5, this particular situation is represented and described by the equation Eq. 4.

$$\partial(t+1) = \delta\partial(t) - E\|J \times \partial_{rabbit}(t) - \alpha(t)\| \quad (4)$$

$$\delta\partial(t) = \partial_{rabbit}(t) - \partial(t) \quad (5)$$

$$J = 2 \times (1 - p_5) \quad (6)$$

Where p_5 is a generated random number from 0 to 1, and jump strength is denoted by J .

(b) **Hard Besiege:** This particular approach enables the potential prey to effectively evade capture while remaining within a state of mild encirclement. When the absolute value of E is greater than or equal to a certain value, and the value of p is less than 0.5, this particular scenario exhibits a higher level of intelligence compared to the one that was previously examined. The levy flight function, as depicted in Eq. 8, is employed to conceptualize this phenomenon.

$$\partial(t+1) = \partial_{rabbit}(t) - E\|\delta\partial(t)\| \quad (7)$$

(c) **Soft Besiege Strategy with Progressive Quick Pounce:**

This particular approach enables the potential prey to effectively evade capture while remaining within a state of mild encirclement. When the absolute value of E is greater than or equal to a certain value, and the value of p is less than 0.5, this particular scenario exhibits a higher level of intelligence compared to the one that was previously examined. The levy flight function, as depicted in Eq. 8, is employed to conceptualize this phenomenon.

$$\partial(t+1) = \begin{cases} Y & \text{if } F(Y) < F(\partial(t)) \\ Z & \text{if } F(Z) < F(\partial(t)) \end{cases} \quad (8)$$

In this case, Y is crucial to the Hawks' subsequent move and may be framed and presented as follows.

$$Y = \partial_{rabbit}(t) - E\|J \times \partial_{rabbit}(t) - \partial(t)\| \quad (9)$$

On certain occasions, the prey will exhibit arbitrary movement in the direction of the raptor. By considering the subsequent LF function, it can be determined that this pattern is a frame.

$$Z = Y + S \times LF(D) \quad (10)$$

$$LF(x) = 0.01 \times \frac{u \times \gamma}{|v|^{\frac{1}{\omega}}}, \gamma = \left(\frac{\Gamma(1+\omega) \times \sin(\frac{\pi\omega}{2})}{\Gamma(\frac{1+\omega}{2}) \times \omega \times 2^{\frac{\omega-1}{2}}} \right)^{\frac{1}{\omega}} \quad (11)$$

u, v are here random numbers that range from 0 to 1, and ω stands for a constant value.

(d) **Hard Besiege: Strategy with Progressive Quick Pounce** This technique depends on the victim not having the strength to escape, and when $|E| < 0.5$ and $p < 0.5$, a harsh besiege is framed over it. This phenomenon is modelled and illustrated in Eq. 12.

$$\partial(t+1) = \begin{cases} Y & \text{if } F(Y) < F(\partial(t)) \\ Z & \text{if } F(Z) < F(\partial(t)) \end{cases} \quad (12)$$

The estimation of Y and Z values are as follows.

$$Y = \partial_{rabbit}(t) - E\|J \times \partial_{rabbit}(t) - \partial_m(t)\| \quad (13)$$

$$Z = Y + S \times LF(D) \quad (14)$$

Algorithm 1 HHO Algorithm

Input: Feature Set

Output: Optimized Feature Set

```

1: Initialize the population  $y_i$  ( $i = 1, 2, \dots, N$ )
2: while condition is not met do
3:   Determine the fitness rate of hawks
4:    $Y_{rabbit}$  is chosen as the best position
5:   for every hawk ( $y_i$ ) do
6:     Update the jump capacity  $J$  and initial energy ( $E_0$ )
7:      $E_0 = 2 \cdot \text{rand}() - 1$ ,  $J = 2 \cdot (1 - \text{rand}())$ 
8:     Update equation using Eq. 3
9:     if  $E \geq 1$  then
10:      Update the position vector using Eq. 1
11:     else
12:       if  $s \geq 0.5$  and  $E \geq 0.5$  then
13:         Update the position vector using Eq. 4
14:       else if  $s \geq 0.5$  and  $E < 0.5$  then
15:         Update the position vector using Eq. 7
16:       else if  $s < 0.5$  and  $E \geq 0.5$  then
17:         Update the position vector using Eq. 8
18:       else if  $s < 0.5$  and  $E < 0.5$  then
19:         Update the position vector using Eq. 12
20:       end if
21:     end if
22:   end for
23: end while
24: return  $Y_{rabbit}$ 

```

The HHO Algorithm 1 technique optimizes hyperparameters to identify the best features. A population of hawks is used as a starting point, and iterative adjustments are made to their placements based on energy and jump height. " Y_{rabbit} ," the best hawk, leads the other hawks. When a hawk has a lot of energy, it uses certain calculations to update its position based on the energy and a random threshold. Until a specific condition is satisfied, this process is repeated to produce optimal features for the proposed model. HHO uses hawks' hunting techniques to identify the best solution.

3.5 CNN

In this novel methodology, CNN (Choudhary et al. 2021; Yu et al. 2017), widely recognized for their expertise in image and text analysis, played a crucial role in the procedure. CNN is specifically engineered to extract both spatial and temporal aspects from data. They have exceptional proficiency in identifying complex patterns and hierarchies within the input. Additionally, CNNs effectively eliminate nonlinearities in the data. These networks have been extensively utilized in several domains, encompassing computer vision, natural language processing, and, specifically, text categorization in this instance. The architectural structure of the CNN is illustrated in Fig. 1.

In this phase, the features that were selected from the HHO are combined with the word embedding. These combined features pass to CNN. The methodology employed in this study entailed the transformation of the feature vectors of news items into two-dimensional matrices. In the following convolutional layer, a set of three separate filters (i.e. 3, 4, 5) was deliberately utilized within the CNN to leverage the computational capabilities of matrix multiplication to extract significant characteristics from the embedding matrix. Each filter smoothly traversed this matrix, resulting in the creation of unique output channels. The use of CNN in this particular scenario enabled the efficient extraction of crucial features from the textual input, hence simplifying the later transition to a BiLSTM network. This significant advancement improved the model's inherent

capacity to identify and understand sequential dependencies and relationships, rendering it highly suitable for various classification tasks.

3.6 BiLSTM

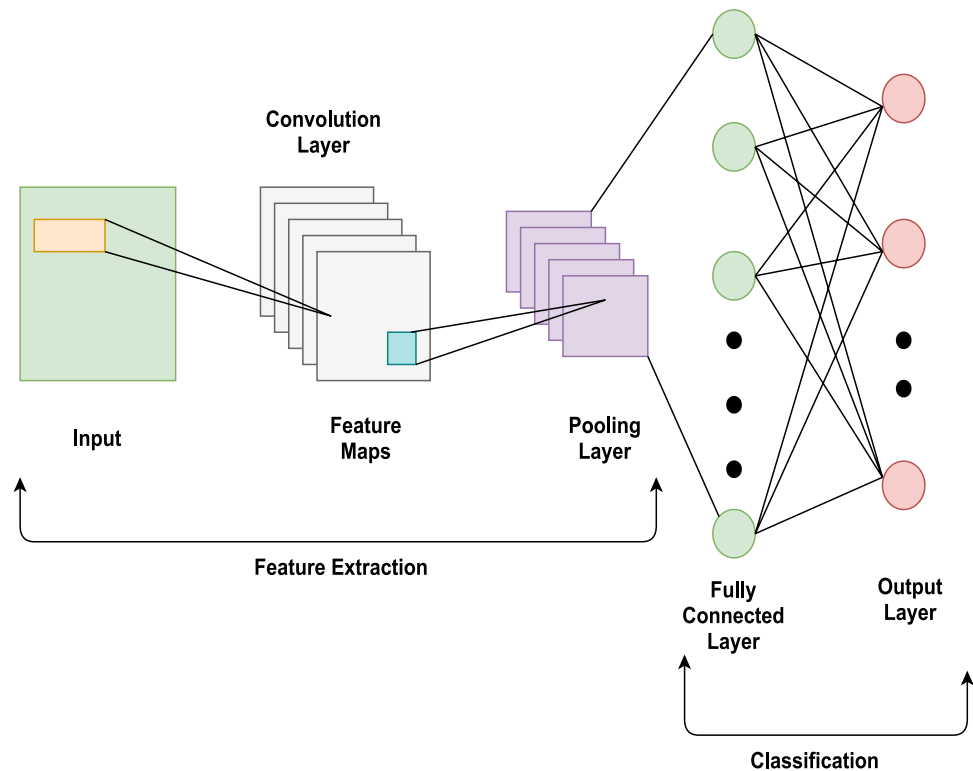
BiLSTM (Choudhary and Arora 2021; Dixit et al. 2022) is an RNN architecture often used in natural language processing and other sequence modeling applications. BiLSTM is an RNN, meaning it is best suited for processing sequential data like language and speech in which the order of the inputs is significant. BiLSTMs, on the other hand, process sequences in both directions at once utilizing two hidden states, as opposed to conventional RNNs which process input sequences in just one way (from start to finish). As the name implies, the forward hidden state works backward via the input sequence, while the reverse hidden state works forward. This makes the BiLSTM ideal for applications like language modeling, machine translation, and named entity identification, where knowledge of the past and the anticipation of the future are equally important.

The formula for a single LSTM cell is:

$$i_t = \text{sigmoid}(W_i * [H_{t-1}, x_t] + b_i) \quad (15)$$

$$f_t = \text{sigmoid}(W_f * [H_{t-1}, x_t] + b_f) \quad (16)$$

Fig. 1 CNN architecture overview



$$y_t = \text{sigmoid}(W_o * [H_{t-1}, x_t] + b_o) \quad (17)$$

$$g_t = \tanh(W_c * [H_{t-1}, x_t] + b_c) \quad (18)$$

$$c_t = f_t * c_{t-1} + i_t * g_t \quad (19)$$

$$H_t = o_t * \tanh(c_t) \quad (20)$$

Here, i_t , f_t , o_t , and g_t represent the input gate activation vector, forget gate activation vector, output gate activation vector, and candidate activation vector, respectively. The cell state vector (c_t) maintains long-term dependencies, while the hidden state vector (H_t) captures the current state information. At each time step t , the input vector (x_t) is processed using weight matrices W_i , W_f , W_o , W_c , which correspond to the input, forget, output, and candidate activations. Additionally, bias vectors b_i , b_f , b_o , b_c are incorporated to adjust the activation functions and enhance learning dynamics. These components collectively enable the model to effectively capture sequential dependencies and retain crucial information across time steps.

The BiLSTM consists of two LSTM networks that process the input sequence in opposite directions, forward and backward shown in Fig. 2. The output of each LSTM network is concatenated to form the final output of the BiLSTM. The formula for the BiLSTM can be represented as:

$$H_t = [FH_t; BH_t] \quad (21)$$

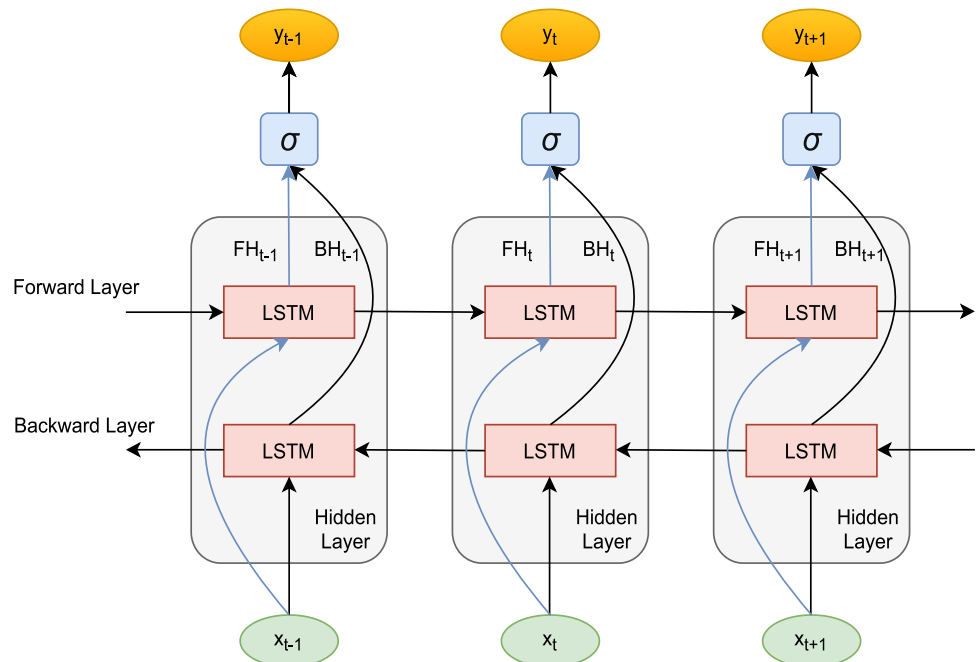
Here, FH_t : output of the forward LSTM network at time t and BH_t : output of the backward LSTM network at time t .

3.7 Proposed hybrid HHO-CNN-BiLSTM model

The HHO method is used to select the optimal features and provide to the hybrid (CNN-BiLSTM) model for classification. The suggested model is shown in Fig. 3. The HHO algorithm, inspired by the cooperative hunting behavior of hawks, was introduced as a nature-inspired meta-heuristic optimization technique. It has demonstrated strong performance in solving complex optimization problems due to its fast convergence and exploration-exploitation balance. Several studies have highlighted HHO's efficiency in achieving high-quality solutions compared to other meta-heuristic algorithms, making it a promising approach for feature selection in machine learning tasks (Heidari et al. 2019; Alabool et al. 2021).

The proposed model flow diagram, Fig. 4, outlines a structured pipeline for dataset collection, preprocessing, feature extraction, optimization-based feature selection, and classification using ML/DL techniques. The process begins with data preprocessing, where raw data is cleaned by handling missing values, removing irrelevant documents, normalizing text, and eliminating noise. Special attention is given to preserving punctuation marks and special characters, as they play a crucial role in fake news detection. Next, feature extraction is performed by deriving eighty LFs using LIWC, self-implemented methods, and various Python libraries. These features capture linguistic, psychological, and quantitative aspects essential for representing the data effectively. In the optimization phase, the HHO algorithm is employed to refine feature selection. Inspired by hawk hunting strategies, HHO

Fig. 2 BiLSTM architecture



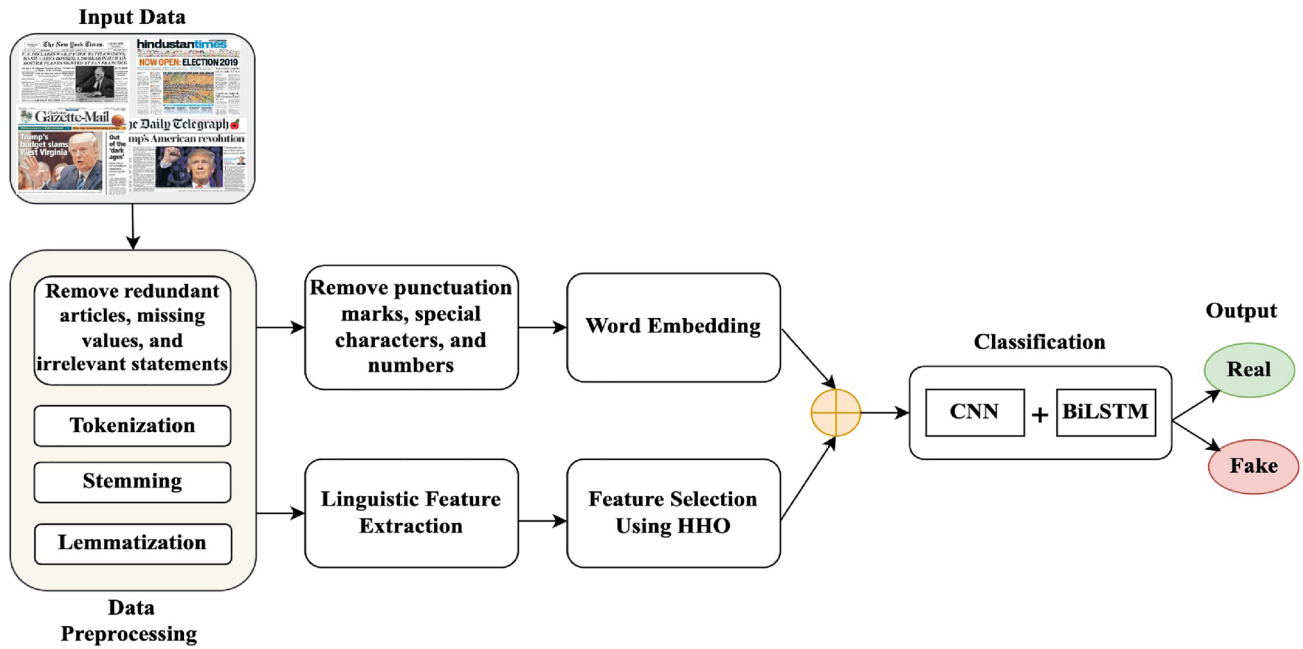


Fig. 3 HHO-based CNN-BiLSTM fake news detection system

initializes a population of hawks and evaluates their fitness using a predefined function. The algorithm iteratively updates hawk positions, searching for the optimal feature subset until convergence is achieved. The selected features are then merged with word embeddings (dimension = 100) and fed into a hybrid CNN-BiLSTM model for classification. The CNN component extracts n-gram features, identifying key textual patterns, while the BiLSTM network processes the sequential dependencies in the text for improved classification accuracy. This structured workflow ensures that the model effectively captures linguistic patterns, selects the most relevant features, and accurately classifies fake news.

Finally, the classification results are evaluated using a confusion matrix, providing insights into performance metrics such as accuracy, precision, recall, and F1-score. This final step concludes the process, offering a comprehensive workflow that spans data preprocessing to advanced classification, all underpinned by optimization and feature engineering.

3.8 Evaluation metrics

In this study, the effectiveness of the hybrid HHO-CNN-BiLSTM model was assessed using performance metrics such as Accuracy, Recall, Precision, and F1 score. These metrics were employed to evaluate the model's performance, and their calculations were based on the provided formulas.

- **Recall:** Recall is the measure of how many true positives were recalled (found), or how many right hits were also located.

$$Recall = \frac{T_P}{T_P + F_N} \quad (22)$$

- **Precision:** The proportion of hits that were truly positive or accurate hits is known as precision.

$$Precision = \frac{T_P}{T_P + F_P} \quad (23)$$

- **F1-Score:** The F1 score, which has a maximum value of 1 and a minimum value of 0, can be thought of as an asymmetrical mean of recall and accuracy. The F1 score is equal parts recall and accuracy. The F1 score is calculated as follows:

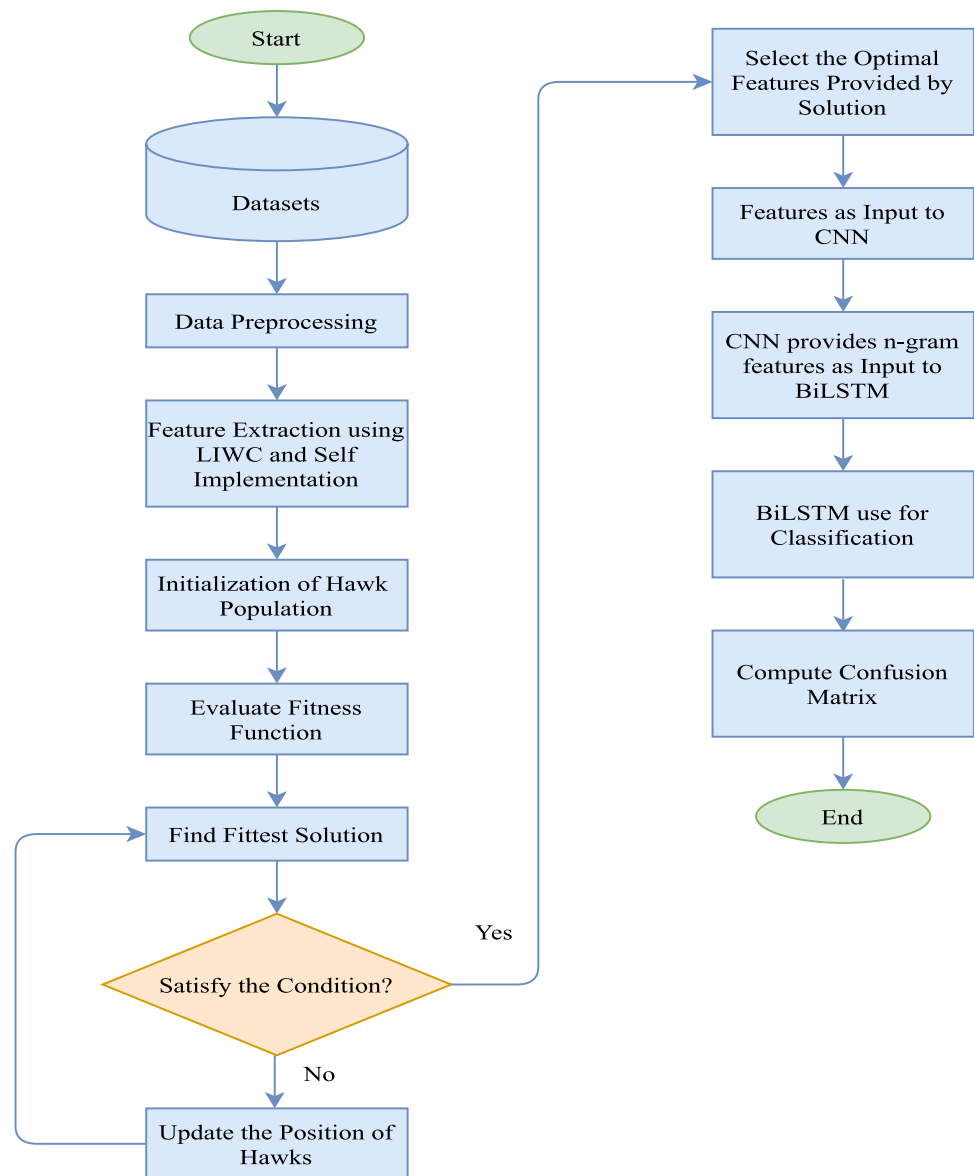
$$F1 - score = \frac{2(Precision * Recall)}{Precision + Recall} \quad (24)$$

- **Accuracy:** The percentage of accurately anticipated values for the test data is used to gauge accuracy. By dividing the total number of forecasts by the total number of accurate guesses, one can easily determine the result.

$$Accuracy = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \quad (25)$$

where, T_P = true positive, F_P = false positive, T_N = true negative, and F_N = false negative.

Fig. 4 Flow diagram of HHO-based CNN-BiLSTM fake news detection system



4 Experiments and discussion

This section describes the methodology used in the experiment with the help of hyperparameter specification, comparison analysis with state-of-the-art methods, and statistical analysis.

4.1 Experimental setup

The proposed model is implemented using Python 3.12 and leverages the Pandas, SciPy, SciKit-Learn, NumPy, Matplotlib, Textblob, Textstat, and NLTK libraries. The complete configuration of hardware and software components is shown in Table 5.

4.2 Hyperparameter specification for CNN-BiLSTM

The performance of the CNN-BiLSTM model is influenced by various hyperparameters, including the number of filters in the convolutional layer, kernel size, dropout rate, learning rate, batch size, and the number of LSTM units. In this study, we manually performed hyperparameter selection based on empirical evaluations to achieve optimal performance. The detailed hyperparameter specifications for the CNN-BiLSTM model are provided in Table 6.

4.3 Results and discussion

This section presents a comprehensive evaluation of the proposed approach for detecting fake news. The analysis encompasses a tabular assessment of diverse performance

Table 5 Hardware and software components and their configurations

		Configuration
Hardware components:	CPU	Intel(R) Core i9 @3.60GHz
	GPU	GeForce GTX 10850
	RAM	64 GB (10 G VENGEANCE)
	Graphics Card	RTX 3080 gaming X TRIO 10 G
Software Components:	Operating System	Window 10 x64
	Tools	Python 3.12

Table 6 Hyperparameter specification for CNN-BiLSTM model

Hyperparameter	Value
# Filter	128
Size of kernel	3, 4, 5
Learning Rate	0.01
Number of 1-D Convolution layer	3 in parallel
Activation Function	ReLU
Max pooling size	2
Number of LSTM Units	64
Dropout Rate	0.5
Number of BiLSTM Layers	1
Activation Function	Sigmoid
Loss function	Binary Cross Entropy
Optimizer	Adam
Batch Size	32
#Epochs	20

metrics, including accuracy, precision, F-1 score, and recall. These metrics are scrutinized across four distinct datasets: ConFake, McIntire, ISOT, and Kaggle dataset. This rigorous evaluation provides a comprehensive overview of the proposed method effectiveness in discerning fake news across various data sources.

The HHO-based feature selection effectively identifies the most discriminative LFs that contribute to fake news detection. Through multiple iterations, HHO consistently selects features related to uncertainty, subjectivity, sentiment, and complexity, as these characteristics are commonly observed in fake news articles. For instance, higher usage of tentative words (e.g., "maybe," "perhaps"), exaggerated punctuation (e.g., excessive question marks and exclamation marks), and readability index are often found in misleading content. Additionally, sentiment-based features, such as negative words (e.g., "fear," "anger") and extreme opinions, play a significant role in distinguishing fake from real news. Another important aspect of HHO algorithm selection process is its ability to reduce redundancy by eliminating less informative or correlated features. This ensures that only the most impactful LFs are retained, improving classification accuracy and reducing computational overhead. The inclusion of stopword frequency, pronoun usage, and unique word

Table 7 Performance of classifiers on ConFake dataset

Classifier	Accuracy (%)	Precision	Recall	F1-score
KNN	77.95	0.77	0.81	0.79
RF	97.31	0.96	0.99	0.97
GNB	82.92	0.84	0.82	0.83
LR	96.21	0.96	0.97	0.96
Bagging	97.09	0.96	0.99	0.97
Adaboost	94.41	0.94	0.95	0.95
SVM	95.08	0.94	0.96	0.95
CNN	97.08	0.97	0.96	0.97
LSTM	96.23	0.95	0.96	0.96
Proposed model	98.56	0.98	0.98	0.98

Bold value represents the higher accuracy/precision/recall/f-1 score rather than other models

count further strengthens the model interpretability, as fake news often relies on persuasive and dramatic language to manipulate readers. By refining the feature space, the proposed model leverages HHO's adaptive searching mechanism to create a more robust and generalizable classification framework, capable of handling evolving fake news patterns across multiple datasets.

4.3.1 Performance of proposed model on ConFake dataset

Table 7 presents the performance of different classifiers on the ConFake dataset. Among ML models, RF (97.31%) and Bagging (97.09%) achieve the highest accuracy due to their ensemble learning capabilities, while LR (96.21%) and SVM (95.08%) also perform well. Adaboost (94.41%) shows competitive performance, benefiting from boosting techniques. However, KNN (77.95%) and GNB (82.92%) exhibit lower accuracy, likely due to their limitations in handling high-dimensional textual data. Deep learning models, CNN (97.08%) and LSTM (96.23%), perform significantly better than most ML classifiers, leveraging their ability to extract spatial and sequential patterns. The Proposed Model (98.56%) surpasses all other methods, achieving the highest accuracy and an F1-score of 0.98, demonstrating the effectiveness of HHO-based feature selection and the hybrid

CNN-BiLSTM architecture in capturing both linguistic and contextual patterns, making it the most reliable approach for fake news detection.

4.3.2 Performance of proposed model on Kaggle dataset

Table 8 presents the performance of different classifiers on the Kaggle dataset using evaluation metrics. Among ML models, LR (96.26%), Adaboost (96.37%), and RF (94.21%) demonstrate strong classification performance due to their ability to capture complex patterns in text data. Bagging (95.60%) also performs well, benefiting from ensemble learning. However, KNN (81.53%) and SVM (81.52%) show lower accuracy, likely due to their sensitivity to high-dimensional feature spaces and potential overfitting to specific patterns. Deep learning models, CNN (96.12%) and LSTM (93.50%), outperform most ML-based classifiers by effectively learning hierarchical and sequential dependencies. The Proposed Model (98.25%) has the highest accuracy and an F1-score of 0.98, demonstrating its superior ability to extract meaningful features and capture both local and contextual dependencies, making it highly effective for fake news detection.

4.3.3 Performance of proposed model on ISOT dataset

Table 9 evaluates the performance of various classifiers on the ISOT dataset. Among traditional ML models, Bagging (97.44%) and RF (96.76%) demonstrate strong classification ability, benefiting from ensemble learning techniques. LR (92.10%), SVM (91.28%), and GNB (87.28%) perform well but show limitations in capturing complex relationships in textual data. KNN (79.34%) records the lowest accuracy, likely due to its sensitivity to high-dimensional feature spaces. DL models, CNN (94.22%) and LSTM (95.33%) outperform most ML classifiers, leveraging their ability to

Table 9 Performance of classifiers on ISOT dataset

Classifier	Accuracy (%)	Precision	Recall	F1-score
KNN	79.34	0.79	0.80	0.79
RF	96.76	0.96	0.96	0.97
GNB	87.28	0.87	0.87	0.87
LR	92.10	0.92	0.92	0.92
Bagging	97.44	0.97	0.97	0.97
Adaboost	95.40	0.95	0.95	0.95
SVM	91.28	0.91	0.91	0.91
CNN	94.22	0.94	0.94	0.94
LSTM	95.33	0.95	0.96	0.96
Proposed model	98.89	0.99	0.98	0.99

Bold value represents the higher accuracy/precision/recall/f-1 score rather than other models

extract hierarchical and sequential patterns from text. The Proposed Model (98.89%) achieves 99% precision and 99% F1-score, which highlights its effectiveness in selecting optimal features and capturing both local and contextual dependencies, making it highly reliable for fake news detection.

4.3.4 Performance of proposed model on McIntire dataset

Table 10 presents a comparative analysis of various classifiers on the McIntire dataset. LR (93.53%) and SVM (93.37%) achieve the highest accuracy, benefiting from their ability to handle high-dimensional data efficiently. RF (92.64%), Bagging (92.85%), and AdaBoost (92.79%) follow closely, leveraging ensemble learning to enhance robustness and reduce overfitting. KNN (87.43%) and GNB (83.27%) show lower accuracy, with GNB suffering from the assumption of feature independence. DL models using word embeddings (WE+CNN: 87.85%, WE+LSTM: 88.32%) perform moderately, demonstrating the ability of CNN and LSTM

Table 8 Performance of classifiers on Kaggle dataset

Classifier	Accuracy (%)	Precision	Recall	F-1 score
KNN	81.53	0.74	0.94	0.84
RF	94.21	0.94	0.94	0.94
GNB	91.09	0.91	0.91	0.91
LR	96.26	0.96	0.96	0.96
Bagging	95.60	0.95	0.95	0.95
Adaboost	96.37	0.96	0.96	0.96
SVM	81.52	0.74	0.95	0.83
CNN	96.12	0.96	0.96	0.96
LSTM	93.50	0.94	0.93	0.93
Proposed model	98.25	0.98	0.97	0.98

Bold value represents the higher accuracy/precision/recall/f-1 score rather than other models

Table 10 Performance of classifiers on McIntire dataset

Classifier	Accuracy (%)	Precision	Recall	F1-score
KNN	87.43	0.88	0.87	0.88
RF	92.64	0.92	0.93	0.93
GNB	83.27	0.85	0.81	0.83
LR	93.53	0.94	0.94	0.94
Bagging	92.85	0.93	0.93	0.93
Adaboost	92.79	0.94	0.92	0.93
SVM	93.37	0.93	0.94	0.94
WE + CNN	87.85	0.89	0.88	0.89
WE + LSTM	88.32	0.89	0.88	0.89
Proposed Model	90.26	0.90	0.91	0.90

Bold value represents the higher accuracy/precision/recall/f-1 score rather than other models

to extract meaningful text features, though they may require more extensive training data for improved generalization. The Proposed Model (90.26%) achieves a strong balance across metrics, benefiting from HHO for feature selection and a hybrid CNN-BiLSTM architecture that captures both spatial and sequential dependencies in text. While it does not surpass the top ML models in accuracy, its feature selection advantage, robust generalization, and high recall (0.91) make it effective for fake news detection, ensuring a reliable trade-off between interpretability and predictive power. The confusion matrix of the proposed model across all four datasets is depicted in Fig. 5.

4.3.5 Comparing proposed method with state-of-the-art

In this comparative analysis, Table 11 evaluates the performance of the proposed method against other state-of-the-art models across four distinct datasets: McIntire, Kaggle, ISOT, and ConFake. To maintain consistency with prior studies (Gravanis et al. 2019; Verma et al. 2021; Choudhary et al. 2021; Dixit et al. 2022; Garg and Sharma 2022), we have used accuracy as the sole evaluation metric for comparing our model with existing approaches. Since all four datasets are balanced, accuracy serves as a reliable and interpretable benchmark for assessing overall classification

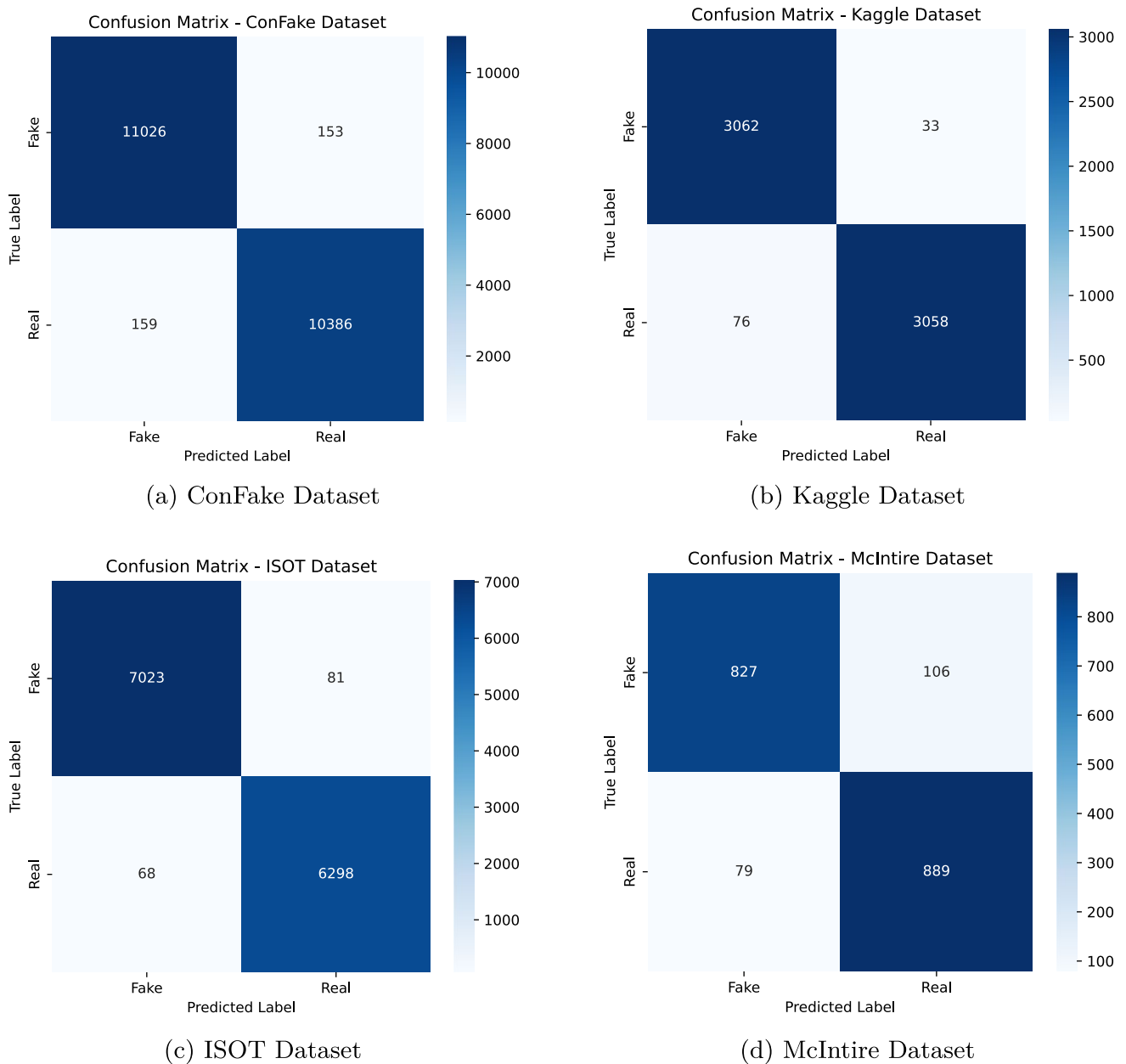


Fig. 5 Confusion matrix of the proposed model on all four datasets

Table 11 Comparison with other related works

Ref.	Dataset(s)	ML model	Best accuracy (%)	No. of LF	Word vector
Kumar et al. (2023)	McIntire, Kaggle, ISOT, and WELFake	OptNet-Fake	98.04	Nil	TF-IDF
Reddy et al. (2020)	McIntire+ PolitiFact	GB	95.49	50	TF, TF-IDF, CBOW, and skip-gram
Gravanis et al. (2019)	Kaggle, McIntire, BuzzFeed, PolitiFact, Unbiased	SVM	95.00	57	Word2Vec
Dixit et al. (2022)	BuzzFeed, PolitiFact, ISOT, GossipCop	PPCA LSTM-LF	94.00	Nil	Latent
Choudhary et al. (2021)	McIntire, Kaggle, GossipCop, PolitiFact	BERT-CNN	97.45	Nil	BERT
Mundra et al. (2023)	McIntire, Kaggle, GossipCop, PolitiFact	CNN, BiLSTM, MLP	95.22	Nil	Ensemble Glove and BERT
Yang et al. (2018)	Kaggle	TI-CNN	92.00	20	Word2Vec
Verma et al. (2021)	McIntire, Kaggle, Reuter, BuzzFeed, WELFake	SVM	96.73	20	TF, TF-IDF
Faustini and Covoos (2020)	McIntire or (Fake_or_real_news dataset)	SVM	94.00	14	BOW, Word2Vec
Choudhary and Arora (2021)	BuzzFeed, PolitiFact	Sequential LSTM	86.00	40	Nil
Kaliyar et al. (2020)	Kaggle	Deep CNN	98.36	Nil	Glove
Proposed work	McIntire, Kaggle, ISOT, BuzzFeed, MD5-Fake	HHO-CNN-BiLSTM	98.89	80	Word2Vec

Bold value represents the higher accuracy/precision/recall/f-1 score rather than other models

performance. Unlike precision, recall, or F1-score, metrics that emphasize specific aspects such as false positives or class imbalance, accuracy provides a straightforward and standardized measure that aligns with previous research. This ensures a fair and direct comparison, objectively demonstrating the superiority of our proposed model.

- **McIntire Dataset:** The McIntire dataset has been evaluated using various models, each demonstrating different levels of accuracy. The BerConvoNet model (Choudhary et al. 2021) achieved an accuracy of 89.92%, while the OptNet-Fake model (Kumar et al. 2023) outperformed it with 92.46% accuracy. Similarly, the PPCA LSTM-LF model (Dixit et al. 2022) attained 89.18% accuracy, whereas the C-BiLSTM model (Trueman et al. 2021) performed slightly lower at 84.22%. The WELFake model (Verma et al. 2021) demonstrated 91.78% accuracy. In comparison, the Proposed Model achieves an accuracy of 90.26%, surpassing most existing models and highlighting its competitive effectiveness in fake news detection.
- **Kaggle Dataset:** On the Kaggle dataset, different models demonstrate varying levels of accuracy. The BerConvoNet model (Choudhary et al. 2021) achieved 96.84% accuracy, while the OptNet-Fake model (Kumar et al. 2023) performed slightly better with

97.46% accuracy. The PPCA LSTM-LF model (Dixit et al. 2022) recorded a lower accuracy of 83.18%, whereas the C-BiLSTM model (Trueman et al. 2021) had the lowest accuracy at 74.14%. The WELFake model (Verma et al. 2021) attained 92.60% accuracy. In comparison, the Proposed Model achieves the highest accuracy of 98.25%, demonstrating its superior performance in fake news detection on the Kaggle dataset.

- **ISOT Dataset:** For the ISOT dataset, different models exhibit varied accuracy levels. The BerConvoNet model (Choudhary et al. 2021) achieved 82.44% accuracy, while the OptNet-Fake model (Kumar et al. 2023) demonstrated a significantly higher accuracy of 98.46%. The PPCA LSTM-LF model (Dixit et al. 2022) recorded 91.28% accuracy, whereas the C-BiLSTM model (Trueman et al. 2021) performed slightly lower with 81.14% accuracy. The WELFake model (Verma et al. 2021) attained 92.04% accuracy. In comparison, the Proposed Model achieved the highest accuracy of 98.89%, reinforcing its effectiveness in fake news detection on the ISOT dataset.
- **ConFake Dataset:** On the ConFake dataset, The Proposed Model maintains its exceptional performance with an accuracy of 98.56%. BerConvoNet (Choudhary et al. 2021), OptNet-Fake (Kumar et al. 2023), PPCA LSTM-LF (Dixit et al. 2022), C-BiLSTM (Trueman et al. 2021)

not perform the experiment on this dataset. WELFake (Verma et al. 2021) achieves 96.73% accuracy, maintaining uniform precision, recall, and F1-score at 0.97 on the WELFake dataset.

4.3.6 Statistical analysis

To validate the effectiveness of the proposed HHO-CNN-BiLSTM model, we conducted a thorough statistical analysis comparing its performance using accuracy with those of baseline models across four benchmark datasets: McIntire, Kaggle, ISOT, and ConFake/WELFake. The baseline models included BerConvoNet (Choudhary et al. 2021), OptNet-Fake (Kumar et al. 2023), PPCA LSTM-LF (Dixit et al. 2022), C-BiLSTM (Trueman et al. 2021), and WELFake (Verma et al. 2021).

The statistical analysis in Table 12 clearly shows that the proposed HHO-CNN-BiLSTM model performs better than most existing models across multiple datasets. The largest improvements are observed on the Kaggle and ISOT datasets, where the proposed model outperforms traditional models like C-BiLSTM (32.49%) and PPCA LSTM-LF (18.12%) on Kaggle, and C-BiLSTM (21.91%) and BerConvoNet (19.96%) on ISOT. The McIntire dataset shows a competitive performance, with a notable gain of 7.18% over C-BiLSTM but a slight underperformance against OptNet-Fake (−2.38%) and WELFake (−1.66%), indicating their effectiveness on this dataset. Similarly, on the ConFake

dataset, the proposed model demonstrates a 1.89% improvement over WELFake and 3.15% over OptNet-Fake, further validating its robustness and adaptability in fake news detection. The overall results confirm that the proposed model achieves better generalization, improved feature selection, and stronger classification capabilities, making it a highly effective solution for detecting fake news.

5 Conclusion and future work

This study presents a comprehensive fake news detection framework, HHO-CNN-BiLSTM, which integrates HHO for feature selection with a hybrid CNN-BiLSTM model for classification. A key contribution of this work is the two-phase data preprocessing approach, which ensures high-quality input for feature extraction and model training. The first phase focuses on linguistic feature extraction, involving duplicate removal, missing value handling, filtering of irrelevant articles, and text normalization through stemming and lemmatization. The second phase refines the data for word embedding extraction, further cleaning the text while preserving semantic elements crucial for misinformation detection. Following preprocessing, eighty LFs are extracted, optimized using HHO for feature selection, and merged with word embeddings for enhanced textual representation. The CNN component captures spatial patterns, while the BiLSTM network effectively processes

Table 12 Relative improvement of the proposed model over existing models

Dataset	Existing Model	Existing model accuracy (%)	Relative improvement (%)
McIntire	BerConvoNet (Choudhary et al. 2021)	89.92	0.38
	OptNet-Fake (Kumar et al. 2023)	92.46	−2.38
	PPCA LSTM-LF (Dixit et al. 2022)	89.18	1.21
	C-BiLSTM (Trueman et al. 2021)	84.22	7.18
	WELFake (Verma et al. 2021)	91.78	−1.66
Kaggle	BerConvoNet (Choudhary et al. 2021)	96.84	1.46
	OptNet-Fake (Kumar et al. 2023)	97.46	0.81
	PPCA LSTM-LF (Dixit et al. 2022)	83.18	18.12
	C-BiLSTM (Trueman et al. 2021)	74.14	32.49
	WELFake (Verma et al. 2021)	92.60	6.11
ISOT	BerConvoNet (Choudhary et al. 2021)	82.44	19.96
	OptNet-Fake (Kumar et al. 2023)	98.04	0.44
	PPCA LSTM-LF (Dixit et al. 2022)	91.28	8.34
	C-BiLSTM (Trueman et al. 2021)	81.14	21.91
	WELFake (Verma et al. 2021)	92.04	7.45
ConFake	WELFake (Verma et al. 2021)	96.73	1.89
	OptNet-Fake (Kumar et al. 2023)	95.55	3.15

Bold values represent the higher accuracy/precision/recall/f-1 score rather than other models

sequential dependencies, leading to superior classification accuracy. The proposed model was rigorously evaluated on four benchmark datasets, ISOT, Kaggle, ConFake, and McIntire, achieving high accuracy rates of 98.89%, 98.25%, 98.56%, and 90.26%, respectively. These results outperform existing state-of-the-art models, demonstrating the effectiveness and generalizability of our approach.

The proposed HHO-based feature selection hybrid CNN-BiLSTM technique has shown efficiency in textual datasets, a key future direction is to explore its applicability to image-based fake news detection, integrating multi-modal data sources. Additionally, the performance of the proposed model needs to be validated on extremely large textual datasets to assess its scalability and generalization

capabilities. Further, more advanced fitness functions can be designed to enhance the selection of the most discriminative features, potentially improving classification accuracy and robustness. These directions will contribute to broadening the applicability of HHO-CNN-BiLSTM model in diverse fake news detection tasks.

Appendix

Table 13 provided the summary of abbreviations and notations used in the paper.

Table 13 Summary of abbreviations and notations used in the paper

Abbreviation/notation	Description
Abbreviations	
ML	Machine learning
DL	Deep learning
CNN	Convolutional neural network
BiLSTM	Bidirectional long short-term memory
HHO	Harris Hawks optimization
LF	Linguistic feature
TF	Term frequency
TF-IDF	Term frequency-inverse document frequency
SVM	Support vector machine
NB	Naïve Bayes
RF	Random forest
KNN	K-nearest neighbors
GB	Gradient boosting
LIWC	Linguistic inquiry and word count
CBOW	Continuous bag of words
T_P, F_P, T_N, F_N	True positive, false positive, true negative, false negative
ReLU	Rectified linear unit
GAN	Generative adversarial network
BERT	Bidirectional encoder representations from transformers
RNN	Recurrent neural network
VGG	Visual geometry group
GNN	Graph neural network
EANN	Event adversarial neural network
LLM	Large language model
KG	Knowledge graph
ISOT	Information security and object technology
NLTK	Natural language toolkit
GRU	Gated recurrent unit
Notations	
$\partial(t+1)$	Position of a hawk in the next generation
$\partial(t)$	Position of a hawk in the current generation
$\partial_{rabbit}(t)$	Position of the rabbit (optimal solution) in the current generation
$\partial_m(t)$	Average position of all hawks at time t : $\frac{1}{N} \sum_{i=1}^N \partial_i(t)$
N	Total number of hawks in the population
p_1, p_2, p_3, p_4, p_5	Random numbers between 0 and 1 for probabilistic modeling

Table 13 continued

Abbreviation/notation	Description
J	Jump strength parameter: $J = 2 \times (1 - p_5)$
E	Escape energy of the prey: $E = 2 \times E_0 \times (1 - t/T)$
E_0	Initial energy of the prey
t, T	Current iteration and maximum number of iterations, respectively
S	Scaling factor for Levy flight function
$LF(x)$	Levy flight function: $LF(x) = 0.01 \times \frac{u \times \gamma}{v^{1/\omega}}$
u, v	Random numbers between 0 and 1 used in Levy flight
ω	Constant governing the Levy distribution
γ	Levy flight parameter: $\frac{\Gamma(1+\omega) \times \sin(\pi\omega/2)}{\Gamma((1+\omega)/2) \times \omega \times 2^{(\omega-1)/2}}$
LB, UB	Lower Bound (LB) and Upper Bound (UB) of the search space

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Data availability The data that support the findings of this study are available at the link: <https://surl.li/bbepth>, Code: <https://surl.gd/xkcgjt>.

Declarations

Conflict of interest The authors declare they have no conflict of interests.

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